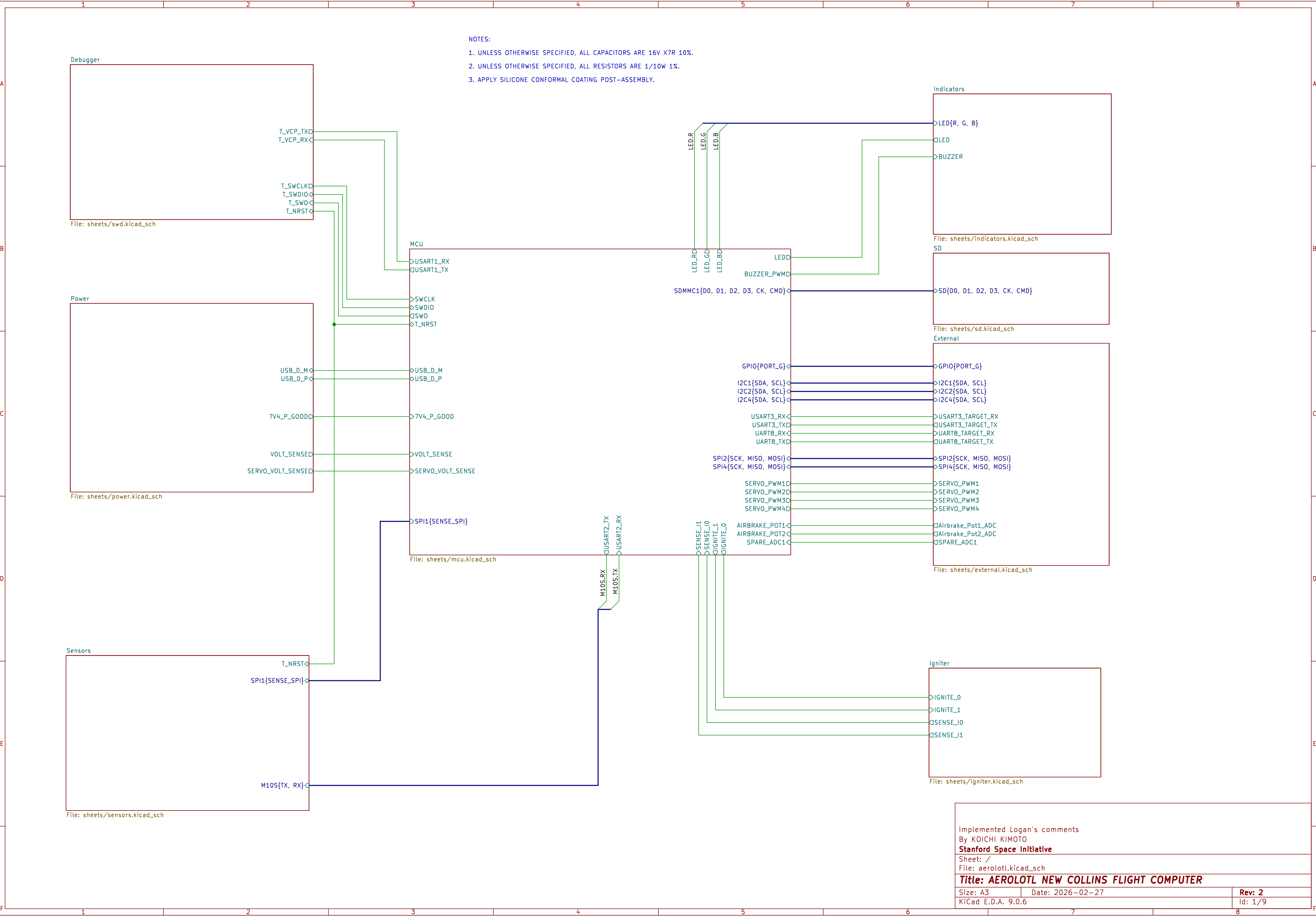
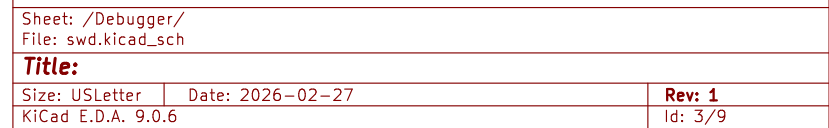






3.3V Supply for MCU

The circuit shows the USB_C_Receptacle_USB2.0_16P (J2) connected to the P_VBUS, USB_DM, and USB_DP pins. The P_VBUS pin is connected to the VBUS pin of the USB shield. The USB_DM and USB_DP pins are connected to the USB_DM and USB_DP pins of the USB shield. The USB shield is connected to the USB_C_Receptacle_USB2.0_16P (J2) and the USB_C_Receptacle_USB2.0_16P (J2) is connected to the USB_C_Receptacle_USB2.0_16P (J2). The USB shield is connected to the USB_C_Receptacle_USB2.0_16P (J2) and the USB_C_Receptacle_USB2.0_16P (J2) is connected to the USB_C_Receptacle_USB2.0_16P (J2).

The 3.3V supply is generated by the AMS1117-3.3 (U4) LDO. The input voltage (Vin) must be greater than the output voltage (Vout) by 1.3V. The output voltage (Vout) is 3.3V. The input voltage (Vin) is connected to the VIN pin of the LDO. The output voltage (Vout) is connected to the VOUT pin of the LDO. The LDO is connected to the USB shield and the USB_C_Receptacle_USB2.0_16P (J2).

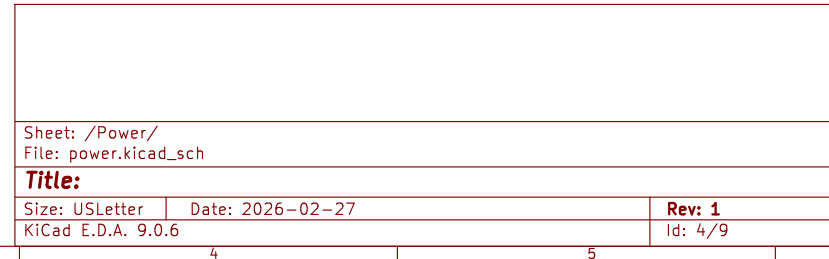
The reverse polarity protection is implemented using the AO4407A (Q4) MOSFET. The MOSFET is connected to the +BATT and +12V pins. The MOSFET is connected to the +BATT and +12V pins. The MOSFET is connected to the +BATT and +12V pins.

12V to 7.4V Stepdown

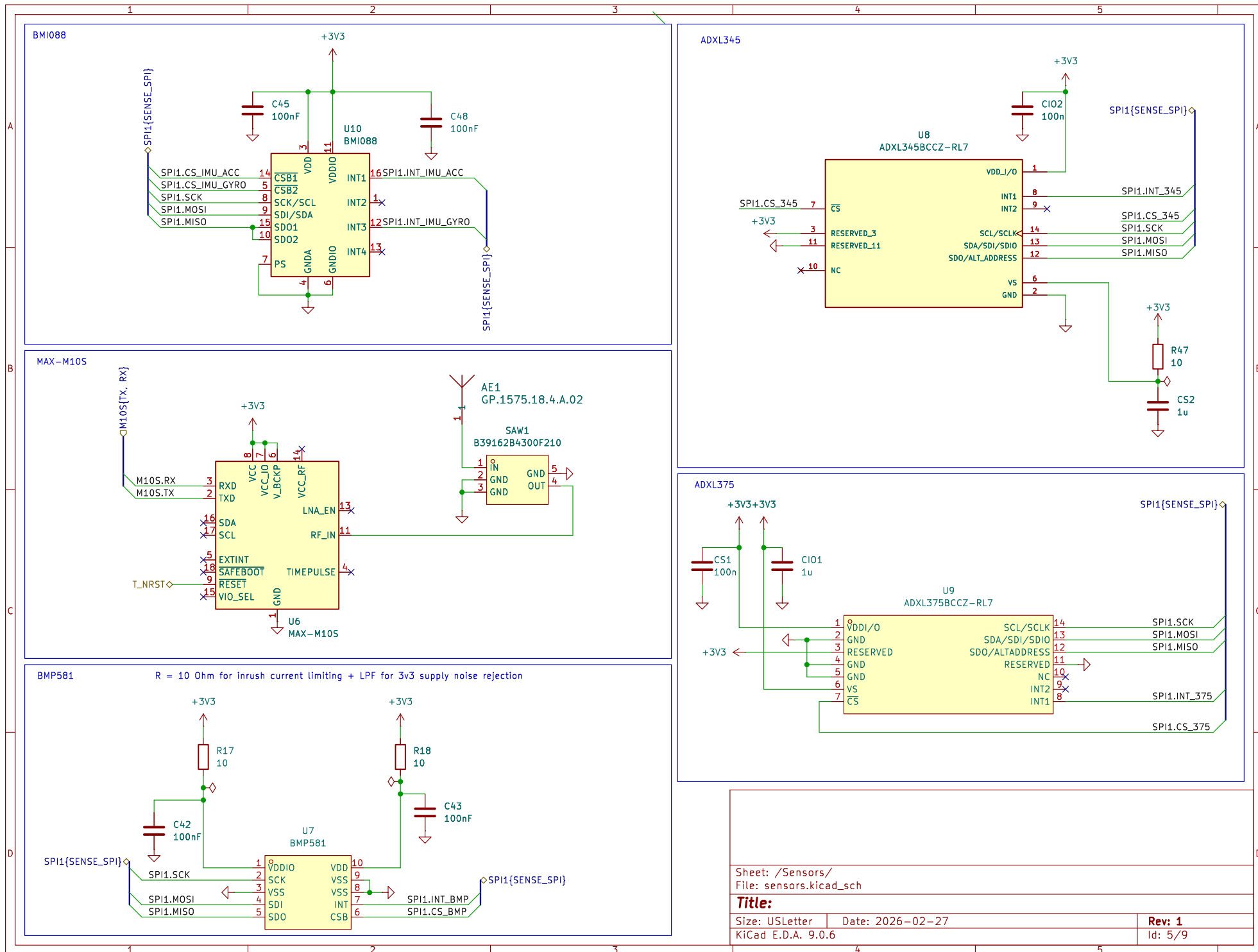
UVLO: $\sim 4.57V$ (rising) / $\sim 4.42V$ (falling)

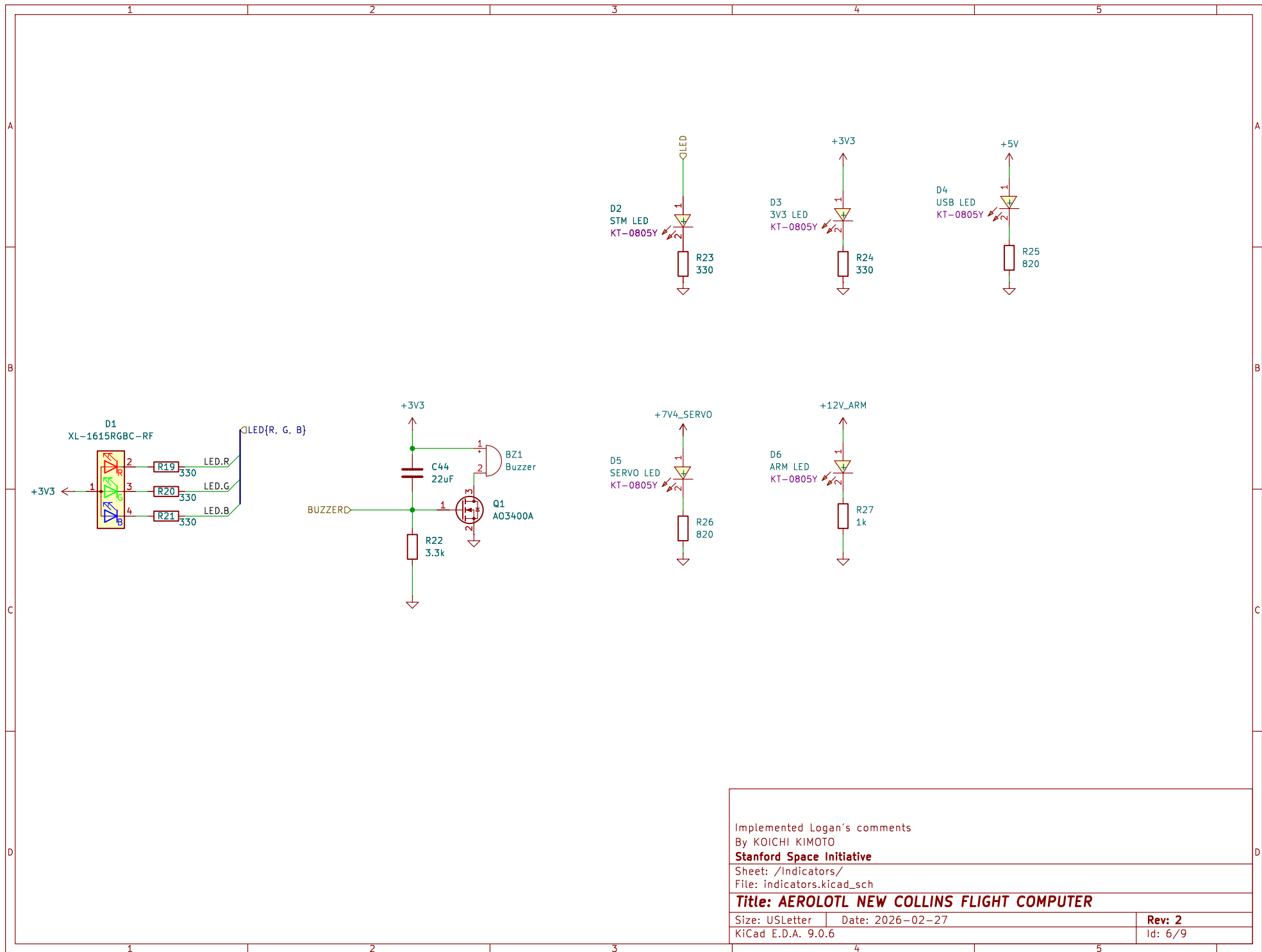
The diagram illustrates a 12V to 7.4V stepdown converter using the TPS54620RGYR IC (U3). The circuit includes the following components and connections:

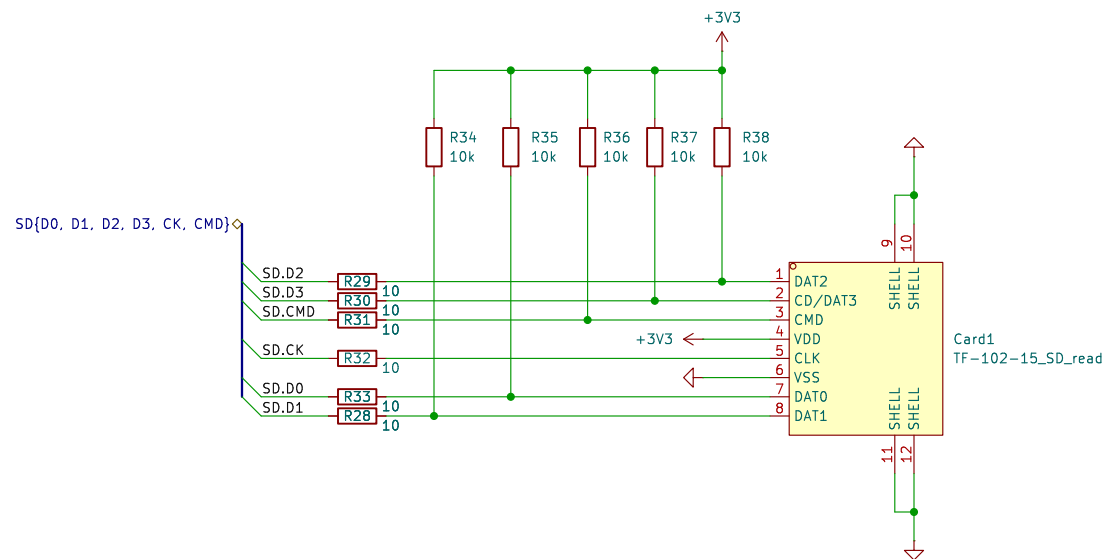
- Input:** +12V supply connected to the VIN pin (pin 5) via a 100k resistor (R5). A 10uF capacitor (C28) is connected to the input.
- IC Pin Connections:**
 - PVIN_4 (pin 4) and PVIN_5 (pin 5) are connected to the input.
 - EN (pin 10) is connected to the input.
 - RT/CLK (pin 1) is connected to the input.
 - SS/TR (pin 9) is connected to the input.
 - GND_2 (pin 2), GND_3 (pin 3), and EXP (pin 15) are connected to ground.
- Output:** The output is taken from the PH_11 (pin 11) and PH_12 (pin 12) pins, which are connected to the output node. A 46uF inductor (L2) is connected to the output. A 7.4V_SERVO supply is connected to the output node.
- Feedback and Compensation:**
 - VSENSE (pin 7) is connected to the output node.
 - COMP (pin 8) is connected to the output node.
 - A 4k3 resistor (R10) is connected between the output node and ground.
 - A 15nF capacitor (C33) is connected between the output node and ground.
 - A 33nF capacitor (C35) is connected between the output node and ground.
 - A 10k resistor (R12) is connected between the output node and ground.
- Other Components:**
 - A 10k resistor (R51) is connected between the output node and the +3V3 supply.
 - A 100k resistor (R6) is connected between the input and the output node.
 - A 56k resistor (R7) is connected between the input and the output node.
 - A 82k5 resistor (R11) is connected between the output node and ground.
 - A 4.7uF capacitor (C29) is connected to the input.
 - A 2n2 capacitor (C30) is connected to the input.
 - A 100nF capacitor (C32) is connected between the output node and ground.
 - A 47uF capacitor (C36) is connected to the output.

[illegible]

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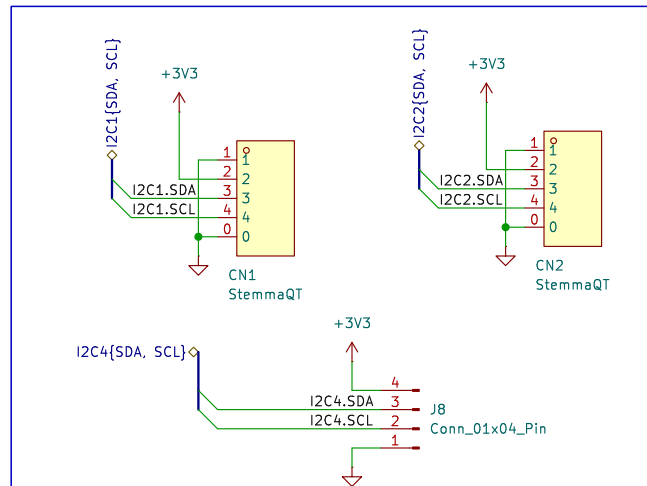




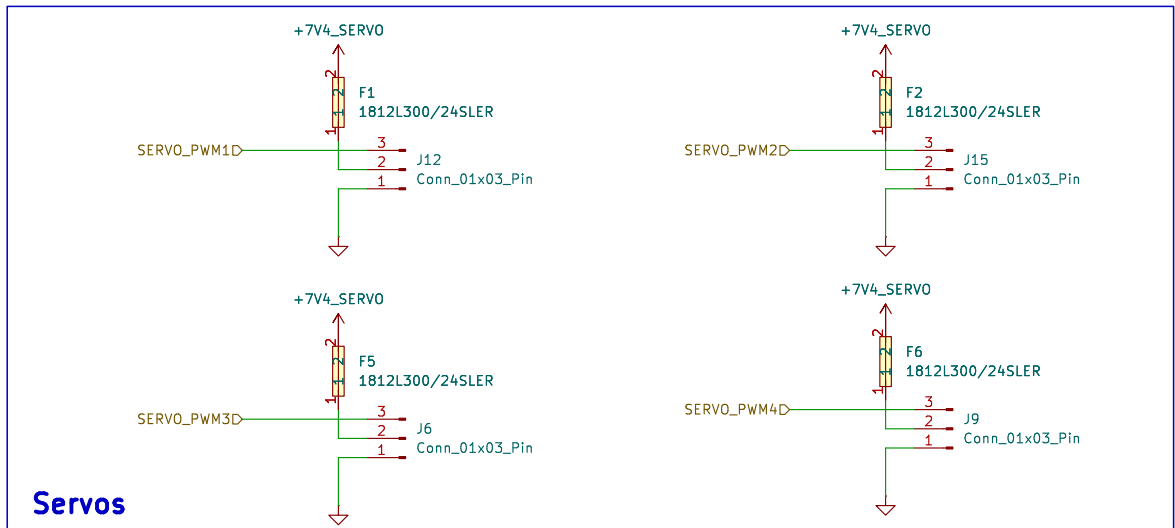


Note: make trace length short (< 2 inch from MCU to sd card reader)

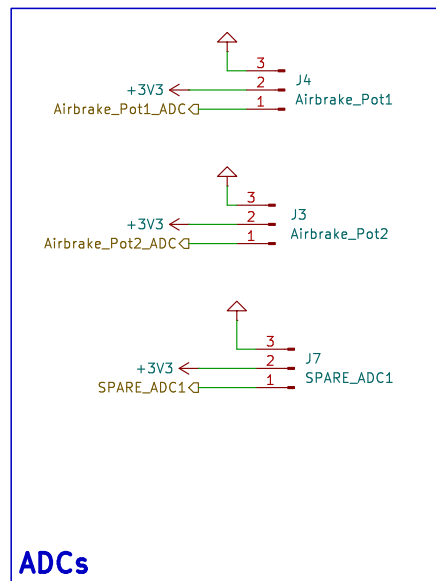
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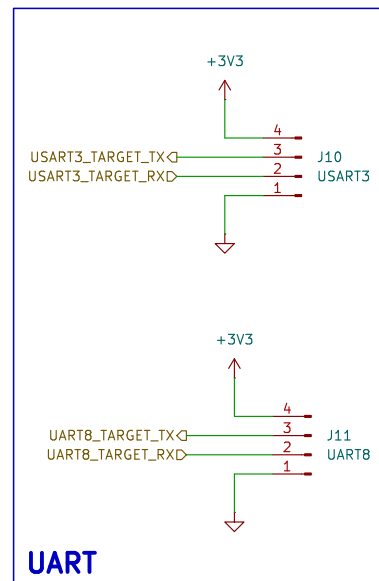
External I2C (two STemma QT, one pinout)



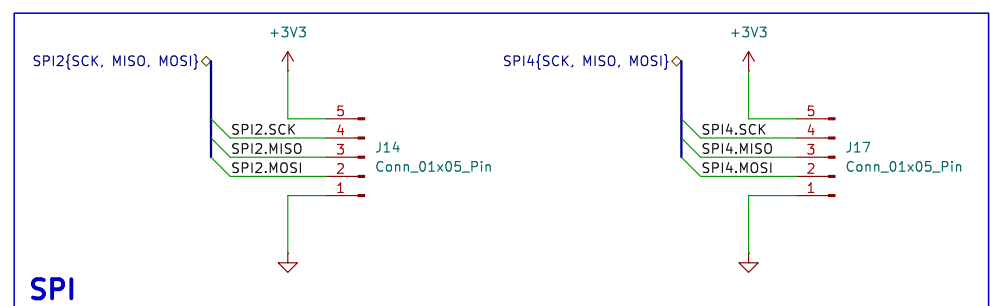
Servos



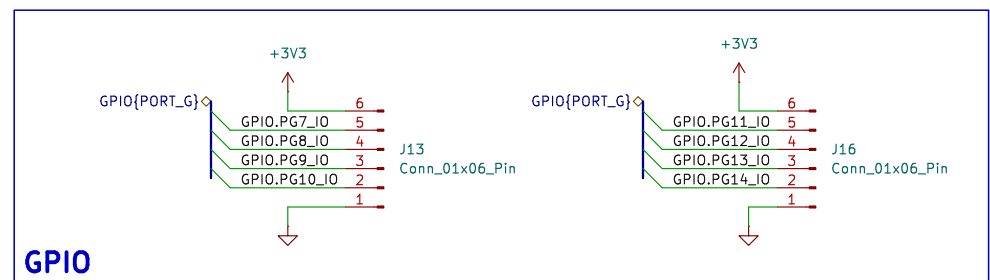
ADCs



UART



SPI



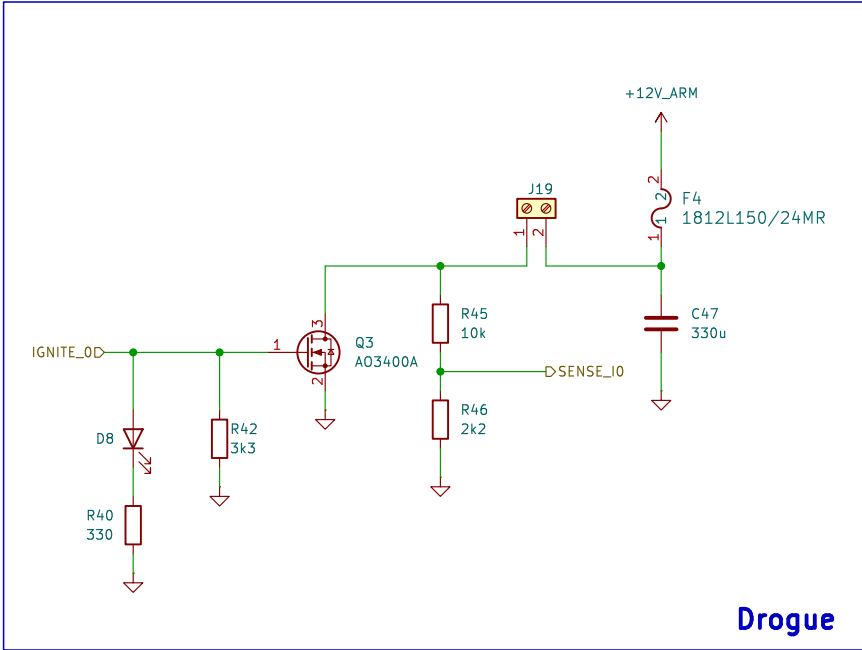
GPIO

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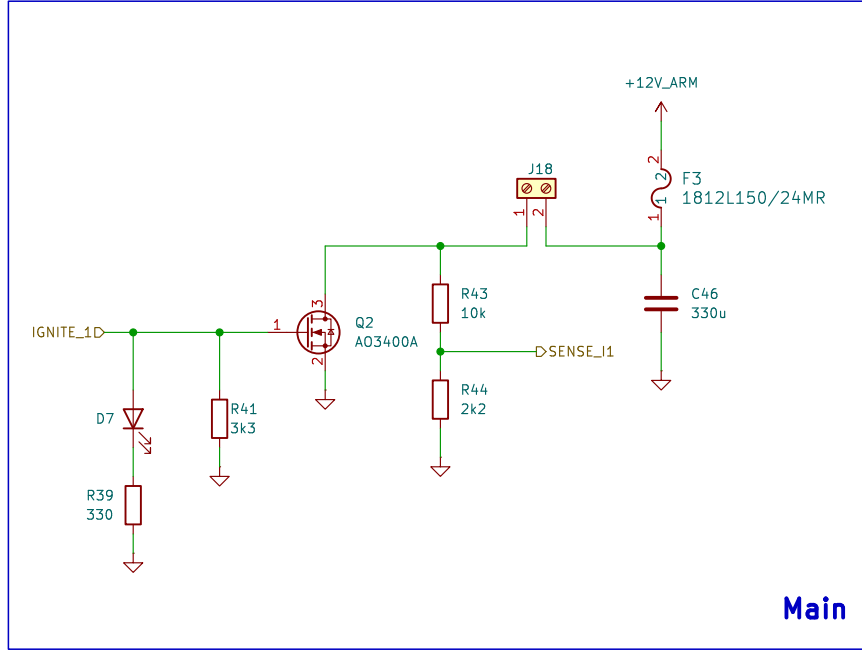
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Droque



Main

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